

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO3_AT2 — DATA ANALYSIS PROJECT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO3 – Analysis (BL4)	Assessment:	Assessment Tool 2 – Data Analysis Project
Weightage:	40% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO3 Statement:	<i>Analyze real datasets using ECDFs, skewed-distribution techniques, Q–Q plots, multi-distribution comparisons, and proportion visualizations to identify patterns and justify conclusions.(BL4)</i>		
Student Name:	SANIA HERBERT	Reg. No.:	192324062
Section / Batch:	2023/AI&DS	Date:	13.8.26

Code of Conduct

I Sania Herbert(192324062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sania

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond construction: analyze the given data, identify the relevant pattern, and justify your conclusion.
- Show all supporting calculations (e.g., IQR, percentiles, angle calculations) where relevant.
- Where a question asks you to compare or recommend, state your reasoning explicitly — a correct chart choice without justification will not receive full marks.

Annexure A – DATA ANALYSIS PROJECT

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT2 — Data Analysis Project

CO3 · BT4: Analyze ·

This rubric is used to assess student responses to the CO3-AT2 Data Analysis Project, in which students analyze a given dataset using ECDF, skewed-distribution, Q–Q plot, multi-distribution, or proportion-visualization techniques and justify their conclusions.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the relevant technique (ECDF, skewed-distribution/log transform, Q–Q plot, multi-distribution comparison, or proportion visualization) and correctly identifies which pattern applies to the given data.	Good understanding of the relevant technique, with only minor gaps in identifying the correct pattern.	Basic understanding, but with some misconceptions about which technique or pattern applies to the data.	Poor understanding of core concepts; misidentifies the pattern or technique required by the data.
Explanation	25%	Provides detailed, well-reasoned analysis explicitly tied to the specific values in the dataset, going beyond description to explain why the pattern occurs or what it implies.	Provides clear analysis with adequate reasoning tied to the data, for most parts.	Analysis is present but generic or only loosely connected to the specific dataset given.	Little or no analytical reasoning provided; answers restate the data without interpretation.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Application	20%	Effectively applies analytical techniques to compare, justify, or recommend a conclusion that is well-supported by the data and any calculations shown.	Applies analytical techniques correctly, with only minor errors in reasoning or calculation.	Limited application; conclusions are only partially supported by the data or calculations.	Fails to apply appropriate analytical techniques; conclusions are unsupported or incorrect.
Organization	15%	The response is clearly structured, addressing every sub-part (a–b) in a logical sequence with coherent overall flow.	The response is organized and addresses each sub-part, with only minor issues in flow or clarity.	Basic structure; sub-parts are addressed but the response lacks clarity or a logical sequence.	Poorly organized; the response is incoherent, and sub-parts are left unaddressed or answered out of order.
Accuracy	10%	All parts of the answer — pattern identification, calculations, and conclusions — are completely accurate and well-suited to the data.	Mostly accurate, with only minor omissions or a small error in one sub-part.	Partially correct; some elements are missing, mismatched to the data, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple sub-parts.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 5

Data: ECDFs of delivery times for two vendors show Vendor A's curve rising more steeply than Vendor

B's specifically in the 20–30 minute range.

Analyze what this local steepness difference indicates about delivery-time concentration in that range.

The ECDF of **Vendor A rises more steeply between 20 and 30 minutes**. This means that a large proportion of Vendor A's deliveries are concentrated within this time interval.

A steep ECDF indicates that **many observations occur in a small range of delivery times**. Therefore, Vendor A has a higher concentration of deliveries around **20–30 minutes** compared with Vendor B.

b) Which vendor would you recommend for time-sensitive deliveries, and justify using the ECDF interpretation.

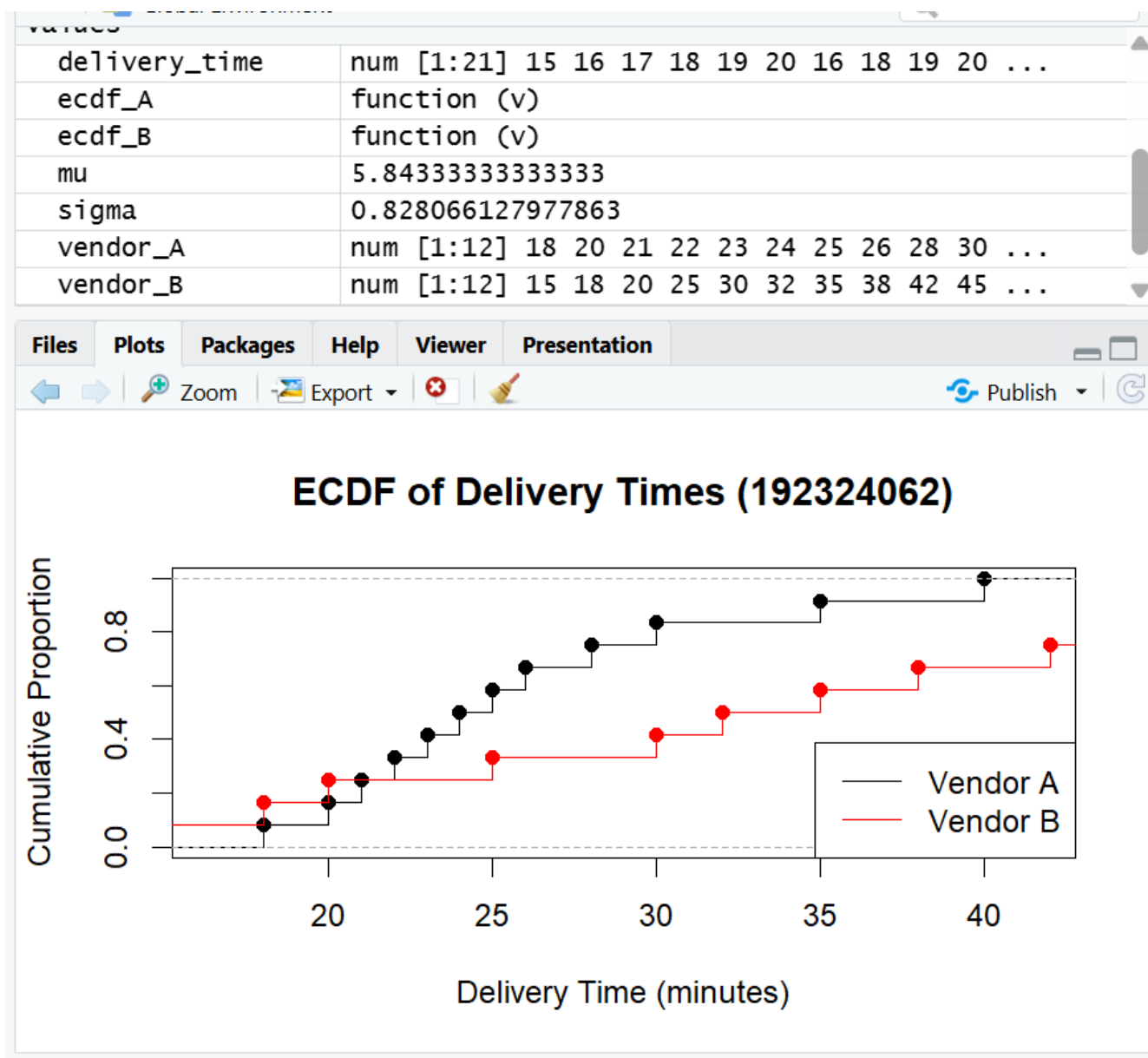
Vendor A would be recommended for time-sensitive deliveries because its ECDF rises more rapidly in the 20–30 minute range. This indicates that a larger proportion of its deliveries are completed within this relatively short time window.

```
7 ecdf_B <- ecdf(vendor_B)
8
9 # Plot ECDF for Vendor A
10 plot(ecdf_A,
11       main = "ECDF of Delivery Times (192324062)",
12       xlab = "Delivery Time (minutes)",
13       ylab = "Cumulative Proportion",
14       verticals = TRUE,
15       do.points = TRUE)
16
17 # Add Vendor B
18 lines(ecdf_B, col = "red", verticals = TRUE, do.points = TRUE)
19
20 legend("bottomright",
21       legend = c("Vendor A", "Vendor B"),
22       col = c("black", "red"),
23       lty = 1)
```

23:16 (Top Level) R Script

Console Terminal Background Jobs

```
R 4.6.1 ~/  
+ ylab = "Cumulative Proportion",  
+ verticals = TRUE,  
+ do.points = TRUE)  
>  
> # Add Vendor B  
> lines(ecdf_B, col = "red", verticals = TRUE, do.points = TRUE)  
>  
> legend("bottomright",  
+       legend = c("Vendor A", "Vendor B"),  
+       col = c("black", "red"),  
+       lty = 1)  
>
```



Problem 6

Data: Monthly income (₹) of 10 households: 15,000, 18,000, 20,000, 22,000, 25,000, 28,000, 32,000, 40,000, 60,000, 500,000.

a) Analyze the shape of this distribution (symmetric, right-skewed, or left-skewed) and justify your answer using the data.

The distribution is strongly right-skewed (positively skewed).

The main reason is the extremely high income of ₹5,00,000, while most households have incomes between ₹15,000 and ₹60,000.

The ₹5,00,000 value is much larger than the other observations and creates a long tail toward the right.

The mean is also much higher than the median:

- Mean = ₹77,000
- Median = ₹26,500

Since mean > median, this supports the conclusion that the distribution is right-skewed.

b) Explain why a log transformation would be appropriate here, and what it would do to the shape of the distribution.

A log transformation is appropriate because the income data contains a very large value (₹5,00,000) compared with the other incomes.

The transformation:

```
log_income <- log(income)
```

compresses large values more strongly than small values.

Therefore, it:

- Reduces the influence of the ₹5,00,000 outlier.
- Shortens the long right tail.
- Makes the distribution less right-skewed.
- Produces a distribution that is closer to symmetric.

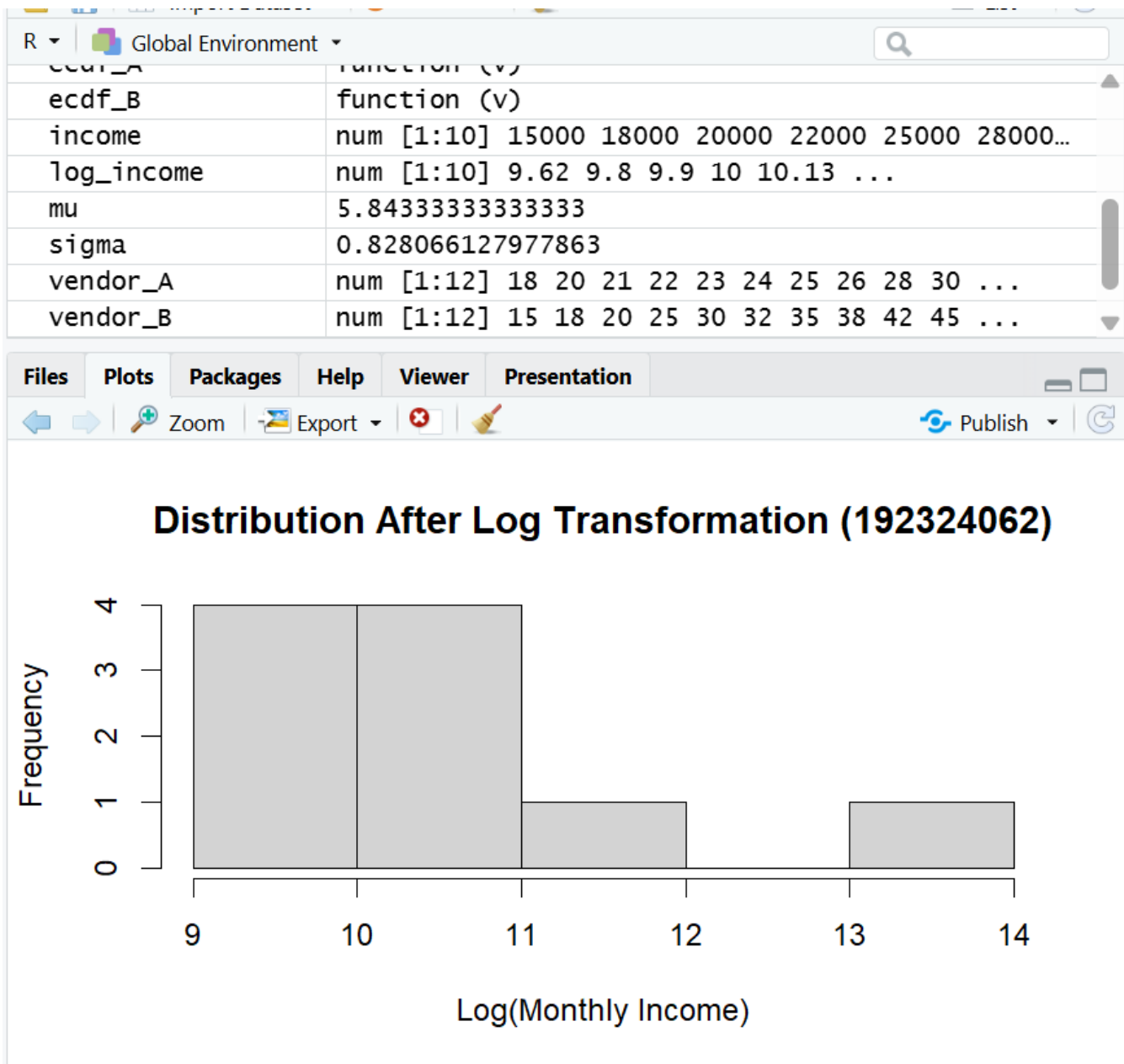
Conclusion: The original income distribution is strongly right-skewed, and a log transformation makes the distribution more balanced and easier to analyze.

```
1 # Monthly income data
2 income <- c(15000, 18000, 20000, 22000, 25000,
3            28000, 32000, 40000, 60000, 500000)
4 # Histogram
5 hist(income,
6      main = "Distribution of Monthly Household Income",
7      xlab = "Monthly Income (₹)",
8      ylab = "Frequency")
9 # Calculate mean and median
10 mean(income)
11 median(income)
12 # Log transformation
13 log_income <- log(income)
14 # Histogram after log transformation
15 hist(log_income,
16      main = "Distribution After Log Transformation (192324062)",
17      xlab = "Log(Monthly Income)",
```

13:26 (Top Level) R Script

Console Terminal Background Jobs

```
> median(income)
[1] 26500
>
> # Log transformation
> log_income <- log(income)
>
> # Histogram after log transformation
> hist(log_income,
+      main = "Distribution After Log Transformation (192324062)",
+      xlab = "Log(Monthly Income)",
+      ylab = "Frequency")
>
```

Problem 12

Data: A Q–Q plot of employee commute times shows a clear “banana” (concave) curve rather than a straight line.

a) Analyze what type of skew this pattern indicates.

A clear **banana-shaped, concave curve** in a Q–Q plot indicates that the commute-time data is **right-skewed (positively skewed)**.

This means most employees have relatively shorter commute times, while a smaller number of employees have **much longer commute times**, creating a long tail toward the right.

b) Explain how this conclusion could also have been reached using a histogram or ECDF, and why the Q–Q plot might reveal it more clearly.

Histogram:

A histogram would show most commute times concentrated on the lower side, with a long tail extending toward higher commute times. This would indicate right skewness.

ECDF:

An ECDF would rise relatively quickly for the common, shorter commute times and then increase more gradually for the few employees with very long commutes. This gradual upper-tail behavior indicates right skewness.

Why Q–Q plot reveals it more clearly:

A Q–Q plot directly compares the observed data quantiles with theoretical normal quantiles. If the data were normally distributed, the points would approximately follow a straight line. The pronounced banana-shaped deviation from the line makes the departure from normality and the direction of skewness easier to identify.

```
1 # Employee commute times
2 commute_time <- c(15, 18, 20, 22, 25, 27, 30, 35, 40, 45, 50, 60, 75, 90)
3 # Q-Q Plot
4 qqnorm(commute_time,
5         main = "Q-Q Plot of Employee Commute Times",
6         xlab = "Theoretical Quantiles",
7         ylab = "Observed Commute Times")
8 qqline(commute_time, col = "pink", lwd = 2)
9 # Histogram
10 hist(commute_time,
11       main = "Histogram of Employee Commute Times",
12       xlab = "Commute Time (minutes)",
13       ylab = "Frequency")
14
15 # ECDF
16 plot(ecdf(commute_time),
17       main = "ECDF of Employee Commute Times (192324062)",
```

21:23 (Top Level) ⚡

R Script

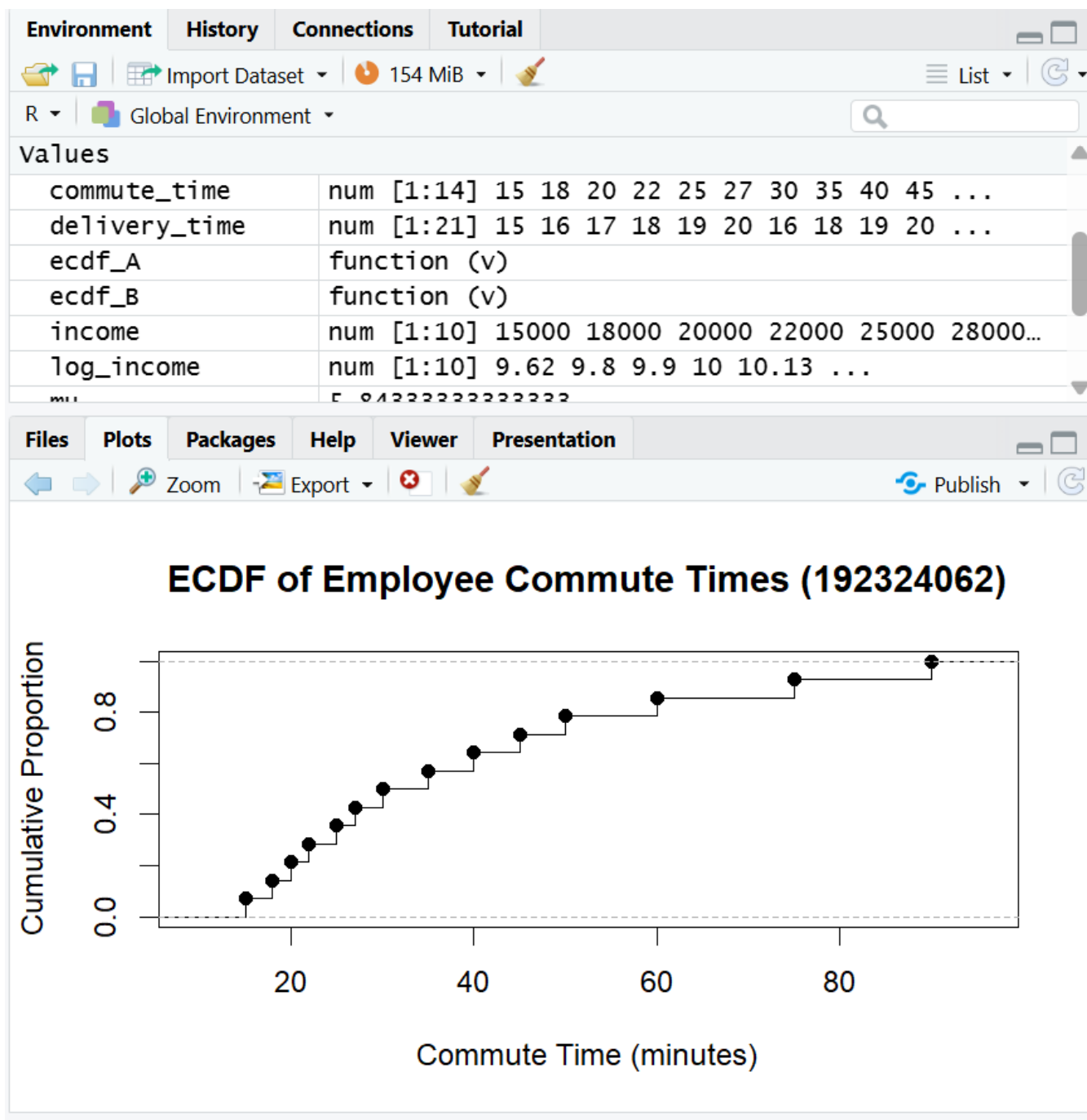
Console

Terminal x

Background Jobs x

R 4.6.1 · ~/

```
+ main = "Histogram of Employee Commute Times",
+ xlab = "Commute Time (minutes)",
+ ylab = "Frequency")
>
> # ECDF
> plot(ecdf(commute_time),
+       main = "ECDF of Employee Commute Times (192324062)",
+       xlab = "Commute Time (minutes)",
+       ylab = "Cumulative Proportion",
+       verticals = TRUE,
+       do.points = TRUE)
> |
```



Problem 18

Data: A bean plot of delivery times shows two distinct bulges — one cluster around 15–20 minutes and another around 45–50 minutes, with a gap in between.

a) Analyze what this bimodal pattern might indicate about the underlying

delivery process (e.g., two different delivery zones or methods).

a) Interpretation of the bimodal pattern

The bean plot shows **two distinct clusters (bulges)**:

- First cluster: approximately **15–20 minutes**
- Second cluster: approximately **45–50 minutes**
- There is a **gap** between the two clusters.

This indicates that the delivery process may consist of **two different groups**. For example:

1. **Nearby delivery zone** → deliveries completed quickly, around 15–20 minutes.
2. **Farther delivery zone** → deliveries take longer, around 45–50 minutes.

Alternatively, the two modes could represent **different delivery methods**, such as normal delivery versus third-party/long-distance delivery.

Therefore, the delivery-time distribution is **bimodal**, suggesting that the data may come from two different underlying processes.

b) Would a box plot alone have revealed this pattern? Explain why or why not.

No. A box plot mainly shows:

- Median
- First and third quartiles
- Interquartile range
- Possible outliers

It does not show the shape of the distribution in detail. Therefore, the two separate clusters and the gap between them may not be visible in a box plot.

A bean plot is better for identifying bimodality because it displays the distribution/density of individual observations.

Conclusion: The bean plot provides more information about the underlying distribution and clearly reveals the presence of two delivery-time groups, whereas a box plot alone may hide this bimodal pattern.

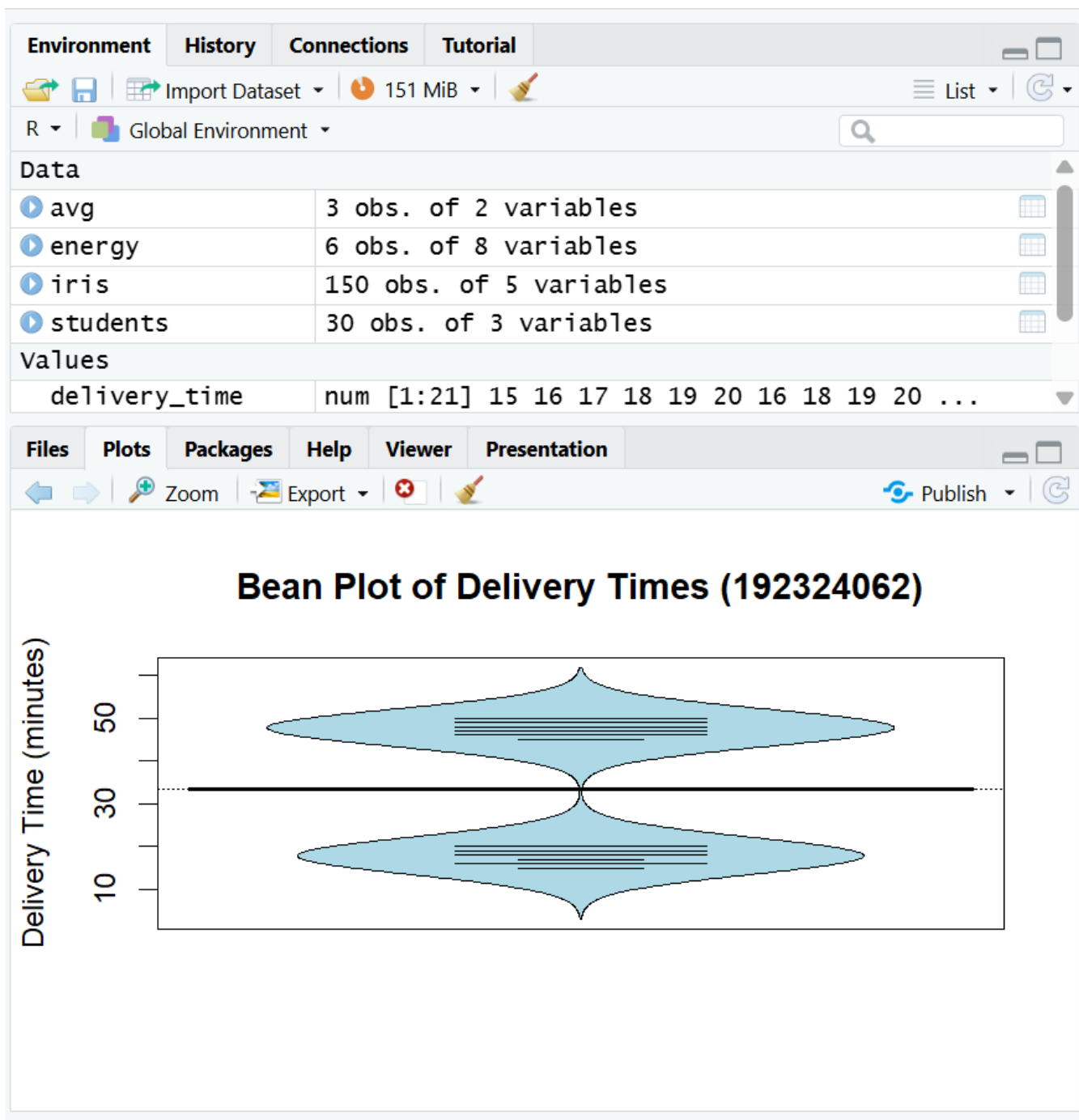
```
1 # Data representing two delivery-time clusters
2 delivery_time <- c(
3   15, 16, 17, 18, 19, 20,
4   16, 18, 19, 20,
5   45, 46, 47, 48, 49, 50,
6   46, 47, 48, 49, 50
7 )
8 # Install package if needed
9 # install.packages("beanplot")
10 library(beanplot)
11 # Create bean plot
12 beanplot(delivery_time,
13           main = "Bean Plot of Delivery Times (192324062)",
14           ylab = "Delivery Time (minutes)",
15           col = "lightblue",
16           border = "black")
```

16:15 (Top Level) R Script

Console Terminal Background Jobs

R 4.6.1 · ~/

```
> # Install package if needed
> # install.packages("beanplot")
>
> library(beanplot)
>
> # Create bean plot
> beanplot(delivery_time,
+           main = "Bean Plot of Delivery Times (192324062)",
+           ylab = "Delivery Time (minutes)",
+           col = "lightblue",
+           border = "black")
>
```



Problem 24

Data: A report states “Department X has 70% women” and separately “Department Y has 40% women,” with no total headcounts given.

a) Analyze why this percentage-only reporting could be misleading, and identify

the critical piece of information that is missing.

The percentages 70% women in Department X and 40% women in Department Y cannot tell us which department actually has more women because the total number of employees in each department is not given.

The critical missing information is the total headcount of each department.

A percentage shows the proportion, not the actual number.

b) Construct a hypothetical example (with totals) where Department Y actually employs more women than Department X, despite its lower percentage.

Suppose:

Department	Total Employees	Women	Percentage Women
X	10	7	70%
Y	100	40	40%

Calculations:

Department X:

So, Department X has **7 women**.

Department Y:

So, Department Y has **40 women**.

Therefore, although **Department Y has a lower percentage of women (40%)**, it actually employs **more women (40)** than Department X (**7 women**).

Conclusion: Percentages should always be considered along with **total headcounts** to avoid misleading comparisons.


```

8 women_Y <- 40
9 percentage_Y <- (women_Y / total_Y) * 100
10 # Display results
11 cat("Department X:\n")
12 cat("Total Employees:", total_X, "\n")
13 cat("Women:", women_X, "\n")
14 cat("Percentage Women:", percentage_X, "%\n\n")
15 cat("Department Y:\n")
16 cat("Total Employees:", total_Y, "\n")
17 cat("Women:", women_Y, "\n")
18 cat("Percentage Women:", percentage_Y, "%\n\n")
19 # Compare number of women
20 if (women_Y > women_X) {
21   cat("Department Y employs more women than Department X.\n")
22 } else {
23   cat("Department X employs more women than Department Y.\n")
24 }

```

24:2 (Top Level) ↕

R Scrip

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R 4.6.1 · ~/

```

> cat("Women:", women_X, "\n")
Women: 7
> cat("Percentage Women:", percentage_X, "%\n\n")
Percentage Women: 70 %

> cat("Department Y:\n")
Department Y:
> cat("Total Employees:", total_Y, "\n")
Total Employees: 100
> cat("Women:", women_Y, "\n")
Women: 40
> cat("Percentage Women:", percentage_Y, "%\n\n")
Percentage Women: 40 %

```